Chi squared goodness of fit tests

1 **D, C, G, F, E, A, B** or **D, C, G, E, F, A, B**

2

a The null hypothesis is that the distribution of students has not changed.
$H_0 : p_1 = 0.55, p_2 = 0.30, p_3 = 0.15, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.55, p_2 \neq 0.30, p_3 \neq 0.15$

b $\alpha = 0.05$

c

	No regular exercise	Occasional exercise	Regular exercise	Total
Observed number of students	113	72	25	210
Expected number of students	115.5	63	31.5	210

d $df = 2$

e $\chi^2 = 2.6811, p = 0.2617$

f $\alpha = 5\%, p > \alpha$, therefore we do not reject the null hypothesis. It appears that there has been no change in exercise levels.

3

a The null hypothesis is that the distribution of smartie colours is even.
$H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = p_8 = \frac{1}{8}, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq \frac{1}{8}, p_2 \neq \frac{1}{8}, p_3 \neq \frac{1}{8}, p_4 \neq \frac{1}{8}, p_5 \neq \frac{1}{8}, p_6 \neq \frac{1}{8}, p_7 \neq \frac{1}{8}, p_8 \neq \frac{1}{8}$

b $\alpha = 0.05$

c 678.5 smarties

d $df = 7$

e $\chi^2 = 72.86$

f $p = 3.89 \times 10^{-13}$

g $\alpha = 0.05, p < \alpha$, therefore we reject the null hypothesis. It appears that the distribution of smartie colours is not even.

4

a The null hypothesis is that the die is fair.
$H_0 : p_1 = p_2 = p_3 = p_4 = p_5 = p_6 = p_7 = p_8 = \frac{1}{8}, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq \frac{1}{8}, p_2 \neq \frac{1}{8}, p_3 \neq \frac{1}{8}, p_4 \neq \frac{1}{8}, p_5 \neq \frac{1}{8}, p_6 \neq \frac{1}{8}, p_7 \neq \frac{1}{8}, p_8 \neq \frac{1}{8}$

b $\alpha = 0.02$

c

	1	2	3	4	5	6	7	8
Observed freq	8	11	12	10	13	11	15	8
Expected freq	11	11	11	11	11	11	11	11

d $df = 7$

e $\chi^2 = 3.64$

f $p = 0.82$

g $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the die is fair.

5

a The null hypothesis is that the city proportion is the same as the school proportion.
$H_0 : p_1 = 0.47, p_2 = 0.37, p_3 = 0.10, p_4 = 0.06, H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.47, p_2 \neq 0.37, p_3 \neq 0.10, p_4 \neq 0.06$

b $\alpha = 0.10$

c

	O	A	B	AB
Observed frequency	88	57	21	6
Expected frequency	80.84	63.64	17.20	10.32

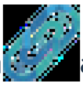
d $df = 3$

e $\chi^2 = 3.97$

f $p = 0.26$

g $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the blood type proportions for the school do not differ significantly from that of the city.

6

- The null hypothesis is that the spinnair. $H_0: p_1 = \frac{25}{36}, p_2 = \frac{1}{6}, p_3 = \frac{5}{36}$
  The alternative is that H_0 is rejected and at least one of the following is true:
  $p_1 \neq \frac{25}{36}, p_2 \neq \frac{1}{6}, p_3 \neq \frac{5}{36}$. The significance level $\alpha = 0.05$.

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	Green	Red	Blue
Observed frequency	142	31	27
Expected frequency	138.89	33.33	27.78

- $df = 2, \chi^2 = 0.25, p = 0.88$
- $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the spinner is fair.

7

- The null hypothesis is that the die is fair. $H_0: p_1 = 0.25, p_2 = 0.5, p_3 = 0.25$
  The alternative is that H_0 is rejected and at least one of the following is true:
  $p_1 \neq 0.25, p_2 \neq 0.5, p_3 \neq 0.25$. The significance level $\alpha = 0.05$.

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	< 4	$4 - 9$	> 9
Observed freq	59	180	61
Expected freq	75	150	75

- $df = 2, \chi^2 = 12.03, p = 0.0024$
- $p < \alpha$ therefore we reject the null hypothesis. It appears that the die is not fair.

8

- The null hypothesis is that the claim is correct.
  $H_0: p_1 = p_2 = p_3 = p_4 = p_5 = \frac{1}{9}, p_6 = p_7 = \frac{2}{9}$
  The alternative is that H_0 is rejected and at least one of the following is true:
  $p_1 \neq \frac{1}{9}, p_2 \neq \frac{1}{9}, p_3 \neq \frac{1}{9}, p_4 \neq \frac{1}{9}, p_5 \neq \frac{1}{9}, p_6 \neq \frac{2}{9}, p_7 \neq \frac{2}{9}$
  The significance level $\alpha = 0.02$.

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	Mon	Tue	Wed	Thurs	Fri	Sat	Sun
Observed freq	25	19	28	22	18	39	45
Expected freq	21.78	21.78	21.78	21.78	21.78	43.56	43.56

- $df = 6, \chi^2 = 3.79, p = 0.71$
- $p > \alpha$ therefore we do not reject the null hypothesis. It appears that the chance of car accidents on the weekend is double that of weekdays.

Critical values

9

a As $\chi^2_{calculated} < \chi^2_{critical}$ we do not reject H_0.

b As $\chi^2_{calculated} > \chi^2_{critical}$ we reject H_0.

10

a The null hypothesis is that the ratings do not change.
$H_0 : p_1 = 0.37, p_2 = 0.25, p_3 = 0.26, p_4 = 0.12,$
$H_1 : H_0$ is rejected and at least one of the following is true: $p_1 \neq 0.37, p_2 \neq 0.25,$
$p_3 \neq 0.26, p_4 \neq 0.12$

b

	Poor	Average	Good	Excellent
Observed frequency	26	67	88	110
Expected frequency	107.67	72.75	75.66	34.92

c $df = 3$

d $\chi^2 = 225.84$

e $\chi^2_{calculated} > \chi^2_{critical}$ therefore H_0 is rejected. It appears the improvements have changed the ratings.